



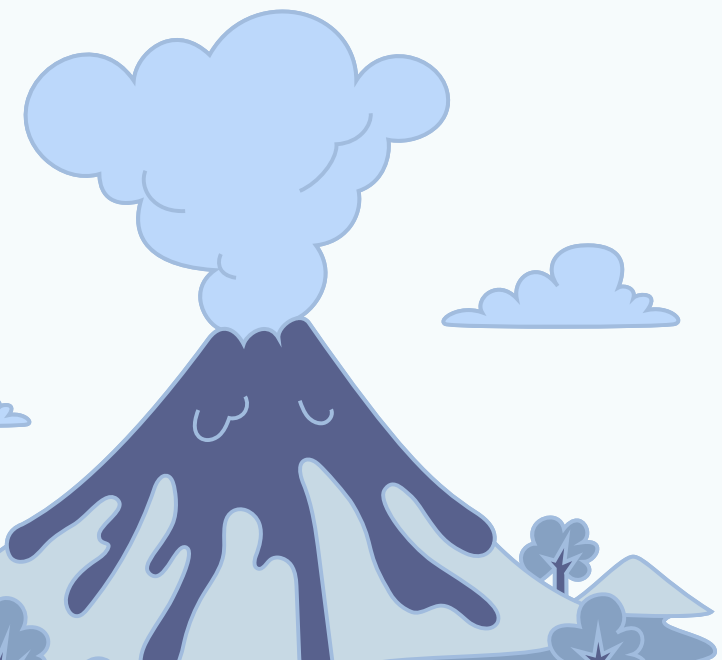
Project AEGIS **(Autonomous Emergency** **Grid &** **Integrated Shelter)**





Problem Statement

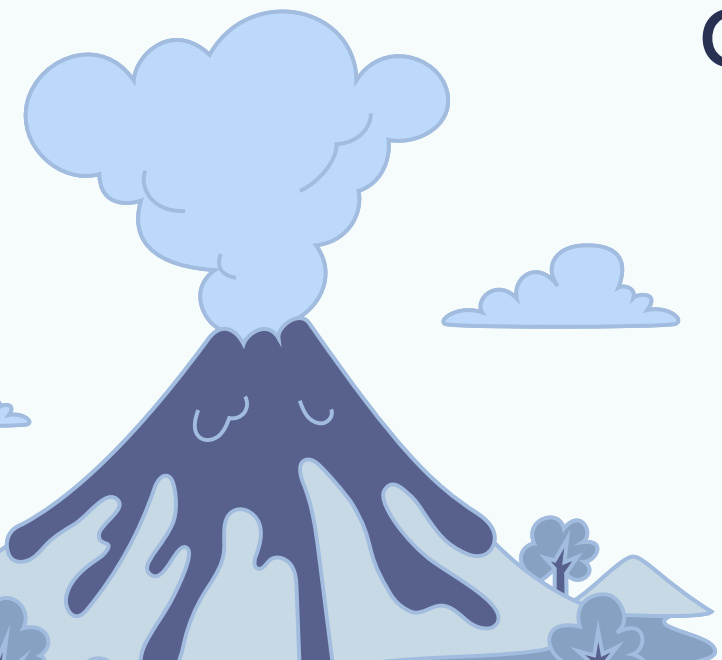
Project AEGIS addresses the critical challenges faced during the early stages of natural disasters such as cyclones, floods, and heatwaves, where electricity, communication, clean water, and safe shelter are often unavailable due to damaged infrastructure.





Overview

A rapidly deployable emergency shelter that becomes operational within 15 minutes, designed to function on uneven terrain and water. It is self-sustaining, providing power, lighting, communication, clean drinking water, and life support during disaster scenarios.





Who, What & Why

Who will use it ?

- Disaster-affected people, who will use the shelter for immediate safety, power, lighting, and basic utilities.
- Government disaster response agencies (such as NDMA, SDRF, NDRF), for rapid deployment during emergencies.
- Humanitarian organizations and NGOs, to support relief camps and early-stage rescue operations.

What pain point are we fixing?

- No immediate safe shelter after disasters
- Lack of power and communication in affected areas
- No real-time monitoring for weather and safety risks
- Limited access to basic utilities like light and clean water
- Slow and inefficient early-stage disaster response

why does the solution matter?

- Provides immediate shelter and essential services during the critical early hours of a disaster
- Works independently of damaged power and communication infrastructure
- Improves safety through real-time monitoring and alerts
- Speeds up rescue and relief operations by enabling quick deployment



Tool Kit

- **Power generation unit**
 - **Energy storage unit**
 - **Microcontroller unit**
 - **Environmental sensors**
 - **Safety sensors**
 - **Alert system**
 - **Communication interface**
 - **Lighting system**
 - **Water storage & purification unit**
 - **Ventilation system**
 - **Structural support & anchoring system**
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- 



Technology used



Control Unit

- ESP32 Microcontroller  (Signal) ->  (Processor)-> 
(Data) ->  (Action)
- Embedded C (Arduino-style programming)
- ESP32 Microcontroller (in case of power leakage unit, this processor will indicate that it is not safe to stay)

Power System

- Foldable Solar panel
- Lithium Battery backup
- Basic charge controller

Safety Sensors

- Smoke / Gas sensor
- Water (flood) sensor
- Tilt sensor



Technology used

Alert System

- Buzzer
- LED indicator

Ventilation

- DC exhaust fan

Display Unit

- LCD display

Structure

- Foldable lightweight frame

Emergency Signaling System

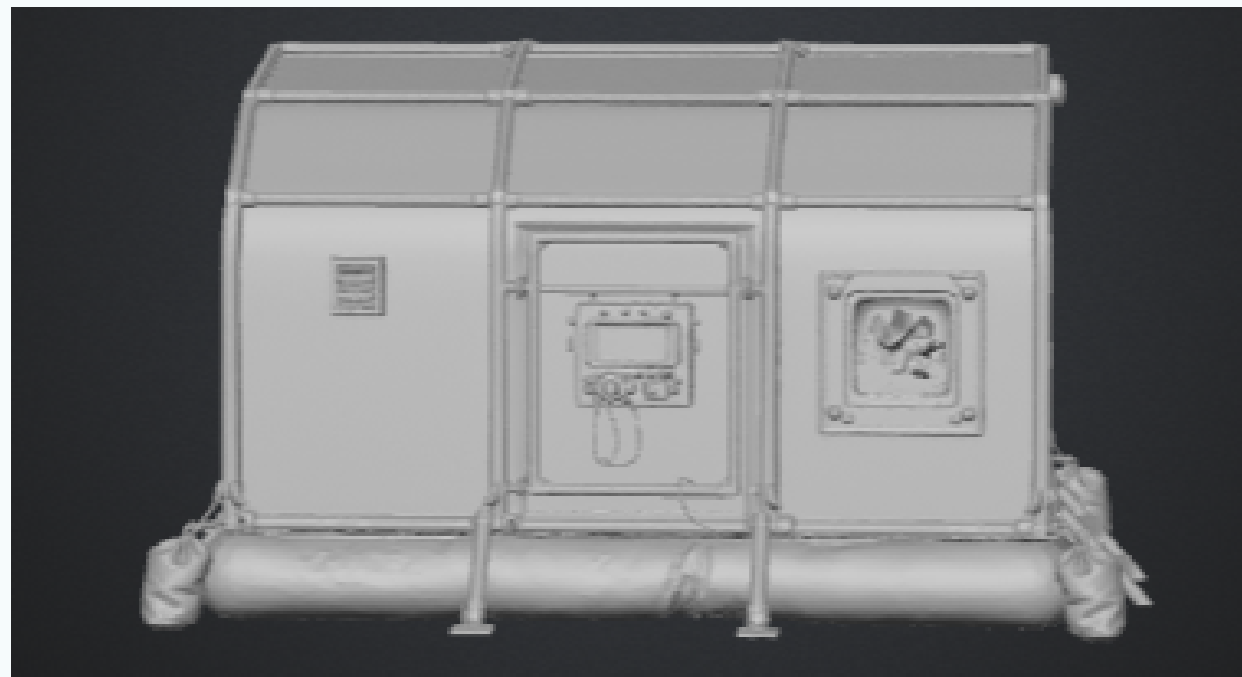
- Morse Code SOS Signaling System (Manual is also available)
- Fastfind(GPS+ Satellite connectivity)



MODEL



Front & Back View

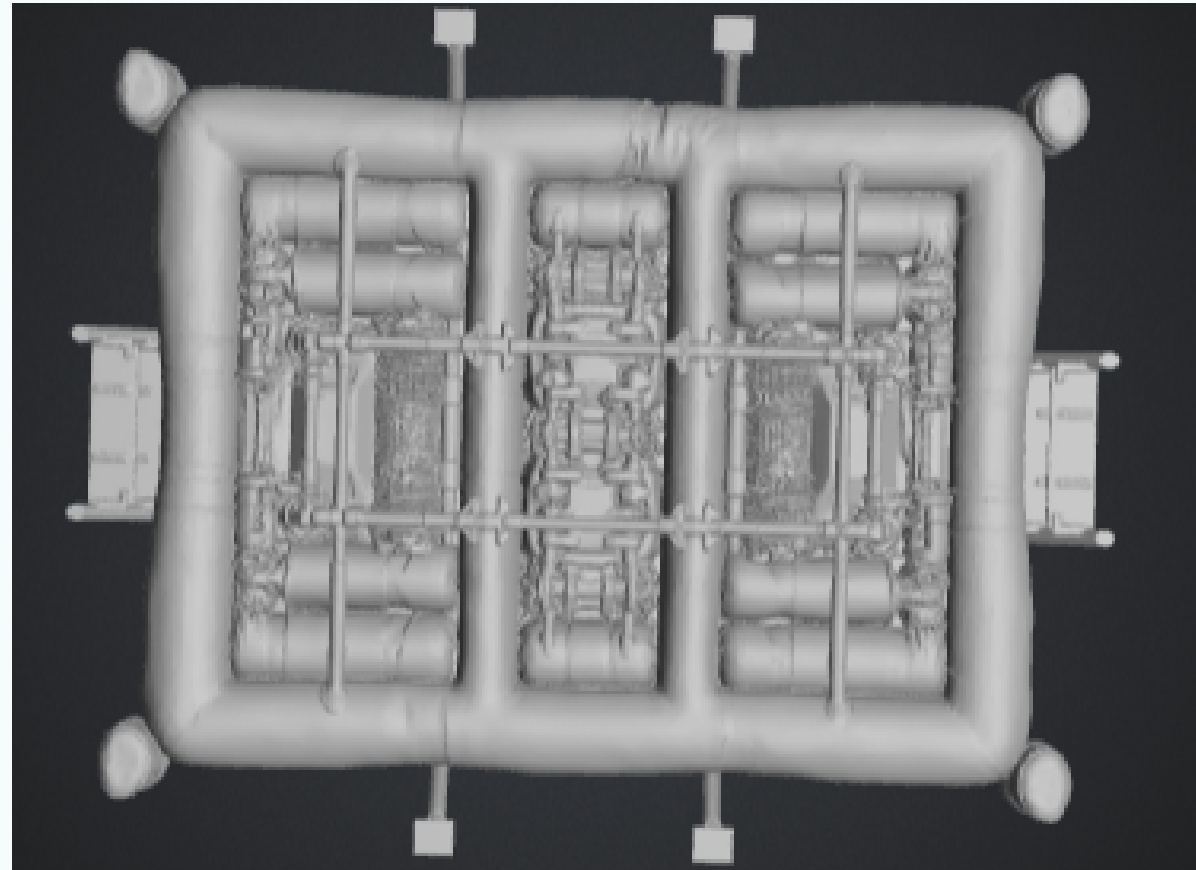


Side view

This view shows the primary entry face of the AEGIS shelter. The sealed front door provides controlled access while maintaining full enclosure against weather and debris. The compact ladder enables safe entry when the shelter is elevated by the inflatable flood base. The arched profile enhances wind resistance and structural stability.

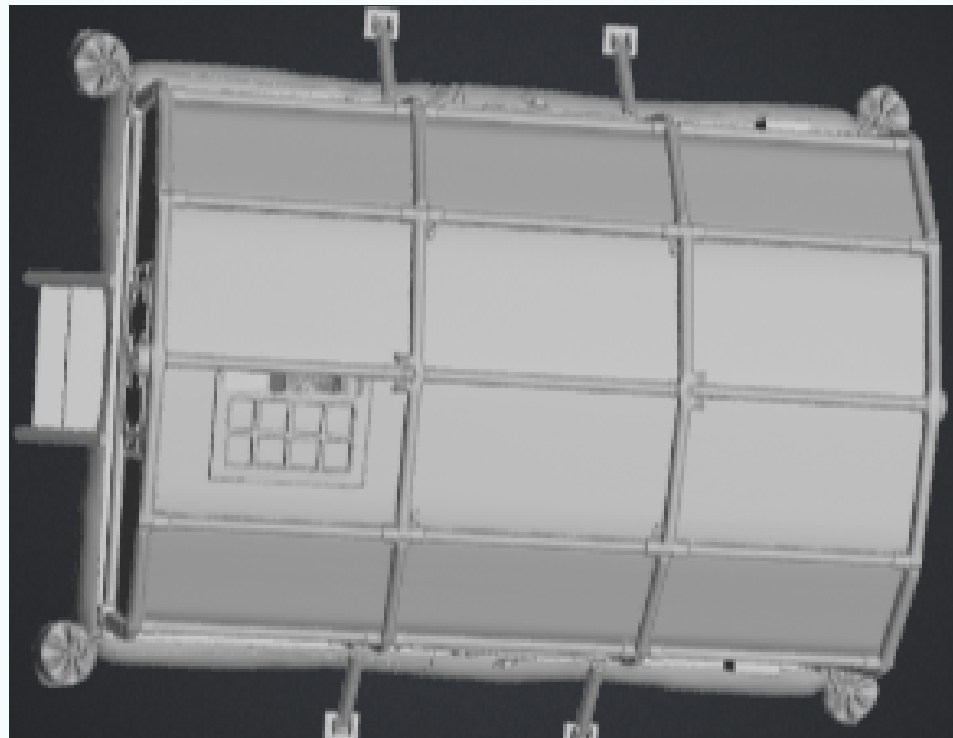
This side highlights the utility and control interface of the shelter. The external access panel houses the electronics system, including the ESP32 control unit, LCD display, and alert indicators, allowing monitoring without opening the shelter. Structural frame members and anchoring supports are visible for lateral stability.

MODEL



Bottom view

This view reveals the perimeter inflatable base system integrated beneath the shelter. The inflatable tubes elevate the structure during flooding, preventing water ingress and improving buoyancy. Internal support elements and anchoring points ensure stability on loose or uneven soil.



Top view

The top view illustrates the tent-inspired segmented roof geometry, designed for efficient rainwater runoff and reduced wind drag. The modular aluminum roof panels improve structural rigidity while allowing easy assembly and transportation. Roof-mounted components such as solar panels can be integrated in this area.

DEPLOYMENT PROCEDURE



AEGIS EMERGENCY SHELTER

Deployment Manual



10–15 min total



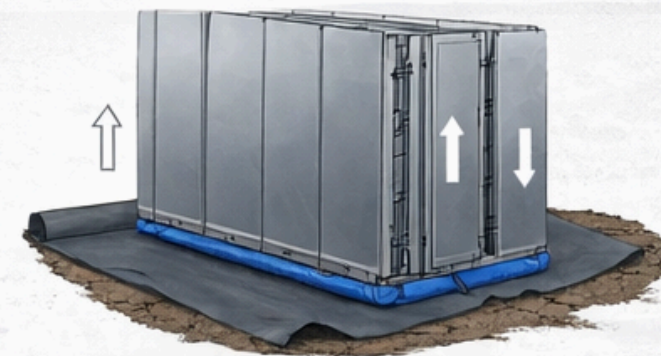
Shelter unit placed on prepared ground



Base sheet laid flat



2–3 operators required



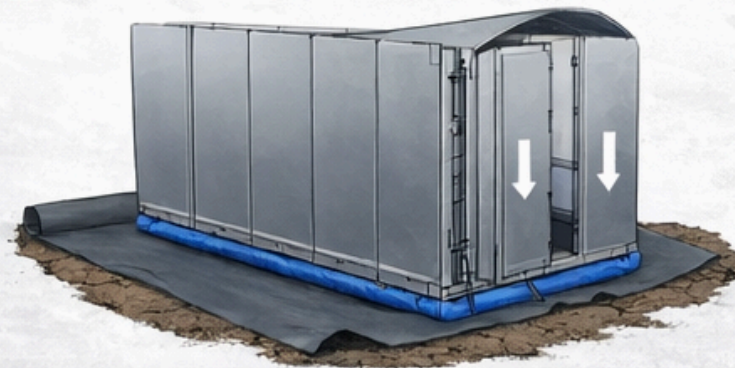
1 Position Packed Unit

Place the compact, fully folded shelter module at the selected site with the base aligned to the ground sheet.

2 Vertical Panel Release

Unlock the vertical locking mechanisms and allow the side panels to extend upward into initial standing position.

4 Roof Extension



5 Stabilize Base

Drive ground spikes through stabilizer legs into the ground and attach ballast bags for extra stability.

6 Connect Power System

Unfold solar panel and connect cables to internal battery control unit.

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7 Secure Shelter Door

Close and seal zippered door fully.





Conclusion

Project AEGIS provides a rapidly deployable, self-sufficient, and highly reliable emergency shelter.

By integrating solar power, intelligent monitoring, and critical utilities, it ensures continuous support, safety, and operational efficiency during extreme weather and disaster events. Its scalable and durable design makes it an ideal solution for early-stage disaster relief, helping save lives and maintain essential services in humanitarian crises.





Thank
You

